Scaffold, Evolve, Then Audit: Cross-Family Re-Judgment Breaks Single-Family Self-Adjudication in an Autonomous Wisdom Loop

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Abstract

Large language models improve markedly when given retrieval-augmented methodological scaffolding: reframes of the problem, anti-patterns to avoid, evaluation criteria, and exemplars of related prior reasoning. A natural next step is an LLM agent that curates its own methodological wisdom library—proposing, validating, and pruning entries without a human in the loop. We build and exercise exactly such a loop, and report a finding that we believe should reshape how self-improvement results in this literature are read.

The setup. A strong scaffolded solver (v20, +28pp vs. the v16 case-library champion on 100 problems, +76pp vs. a budget-matched baseline on a 50-problem held-out set) is paired with four orchestration modules: a failure generator, a success distiller, a cross-model distiller, and a Darwinian pruner. Each candidate wisdom must pass a +10pp held-out A/B gate to enter the library.

The inner-loop result looks strong. Over a single cycle the library grows $75 \to 77 \to 58_{\text{active}} + 19_{\text{deprecated}} \to 59_{\text{active}} + 19_{\text{deprecated}}$. Three new wisdoms are committed ($W_{076}, W_{077}, W_{078}$) from disjoint evidence sources; nine candidates are refused; nineteen stale entries are deprecated. Only 3/12 proposals pass the gate—evidence that the A/B gate is doing real work.

The outer-loop audit breaks the result. We then re-judge the same 150 answer pairs (50 held-out problems $\times$ 3 KEEP decisions) with a different LLM family (Claude Opus 4.6). 0/3 KEEP decisions survive cross-family re-judgment: W_{076} falls from $0.64 \to 0.40$, W_{077} from $0.60 \to 0.47$, W_{078} from $0.60 \to 0.51$. Every single KEEP flips to REVERT under the cross-family judge. A self-rebuttal step, run before the cross-judge was invoked, had already predicted this: the agent returned `further_test` (never `accept`) on all three KEEPs, naming judge-preference alignment as the strongest confound. The cross-judge confirmed the rebuttal.

The take-away. Single-family self-adjudication is not enough. An agent can be simultaneously (i) honest about its gate (9/12 REVERTs, disjoint evidence sources, auditable logs), (ii) successful by its own measurement, and (iii) wrong in a way that only becomes visible once a separate-family judge re-scores the identical text. We argue cross-family re-judgment should become a routine reporting item for any self-improving LLM loop, not a stretch goal.

We then close the recursion. A final experiment (`auto_recurse.py`, §4.7) lets the agent chain audit $\to$ belief-update $\to$ gate-proposal $\to$ re-audit by itself. It converges in three cycles to a strictly stricter fixed-point gate (`gemini_wr`$\geq$`0.60 AND claude_wr`$\geq$`0.55`) and, under that gate, withdraws every one of its own previous commits.

We then pull the original plan back in. A staged validation funnel (pilot at n=10, comparative at n=50, cross-domain, cross-family) run retrospectively on all 12 candidates produces $12 \to 4 \to 0 \to 0 \to 0$: zero candidates survive. The unexpected result is that even same judge family, same content, different side-randomization seed suffices to flip every single original KEEP decision. The gate is fragile along three independent axes simultaneously: seed, family, and domain aggregation.

Triangulation is not enough. Having falsified the single-family gate, we bring in a third LLM family (GPT-5.4) and run majority-of-3 voting at $n = 50$. This gate looked promising—it rejected two of the three original KEEPs (W_{076}, W_{077}) as family-bias false positives, confirmed the third (W_{078}) by 2/3 family agreement, and rescued a REVERT (WCAND07). We then ran the agent's own meta-wisdom falsifier: extend the three KEEPs to $n = 100$ on fresh problems. At $n = 100$ two of three stabilise at $\text{wr} = 0.52 < 0.55$ (the meta-wisdom's prediction holds); the third stays in the gray zone at 0.57. Zero candidates hold above 0.60 at $n = 100$. The majority-of-3 verdict at $n = 50$ that survived for W_{078} collapses at $n = 100$: gemini drops from 0.60 to 0.52, claude from 0.51 to 0.49. Final autonomous library

delta, after every audit: $\boxed{+0}$ KEEPs; the loop's committed wisdoms are all either family-bias FPs or noise-driven at $n = 50$.

Two orthogonal measurement axes arrive at the same failure mode from outside pair-wr. Embedding-space faithfulness (sentence-transformer cosine between perturbation-direction and wisdom-description) is near-zero for all 12 candidates, and LLM-judged faithfulness (Claude over 20 pids per candidate) never reaches a strict-YES rate above 0.10. The candidates cause real perturbations to the solver's Turn-0 reasoning (perturbation magnitudes in $[0.71, 0.77]$), but the direction of perturbation is not the direction the wisdom self-advertises. Put differently: pair-wr measures generic context-perturbation lift; it does not measure whether the specific wisdom is doing the work. This is independent evidence for the pair-wr collapse from a measurement frame with zero data-overlap.

Contribution. The loop, the ten-experiment audit stack (cross-family, side-stability, cross-domain, staged funnel, recursion-to-fixed-point, triangulation, inverted- disagreement check, meta-wisdom's own falsifier, embedding-space orthogonal validation, LLM-judged faithfulness), the multi-dimensional failure modes they reveal across sample size, judge family, judge seed, domain aggregation, and direction-of-activation, and the agent's meta-wisdom independently predicting the extension result. A concrete step toward agents that build the tools they think with and notice when their own tools mislead them—even if the notice costs all three of their provisional commits.

1 Introduction

A large language model asked to "just solve it" will typically underperform a version of itself given a well-matched cognitive prosthesis: a problem reframe, a list of anti-patterns, the memory of two superficially different but structurally parallel prior cases, and an evaluation rubric. The gap between the raw solver and the scaffolded solver is usually attributed to chain-of-thought [Wei et al., 2022], retrieval-augmented generation [Lewis et al., 2020], self-reflection and verification [Madaan et al., 2023, Shinn et al., 2023], or agentic decomposition [Yao et al., 2023b]. We call the artifact that these methods all point to a wisdom: a small piece of methodological knowledge of the form "in situations exhibiting pattern P, the right move is M; the canonical failure mode is F."

A working wisdom library is a retrieval corpus of such entries, selected per problem, foregrounded in the prompt, and paired with cross-domain exemplars that show what the wisdom looks like when instantiated in a specific domain. In our setting (see §3), we use 75 such wisdoms, each authored in the classical tradition (aphorisms from Yi Jing, Munger, Drucker, Mao, Confucius, folk proverbs, etc.) and unpacked into LLM-legible form. Over six iterative framework revisions (v13 through v20, see Figure 2), such a scaffolded solver rises from 74% to 86% win rate against a budget-matched baseline.

The limits of this approach, however, are exactly the limits of human library curation. Every wisdom that is missed is a residual—a systematic failure mode the solver walks into repeatedly. Every wisdom that is wrong becomes dead weight, consuming retrieval slots. And every wisdom that is nearly right but poorly phrased costs the solver one full pass of the scaffold.

This paper asks a direct question: can the agent itself curate its wisdom library? Concretely, we ask whether an LLM agent, running under a reasonable scaffold, can:

(i) Detect recurrent orientations in its own reasoning that a static library has not yet articulated;

(ii) Propose new wisdom entries with source, signal, unpacking, and exemplars;

(iii) Filter proposals against a held-out A/B gate that is strict enough to refuse most of them;

(iv) Prune stale wisdoms that have stopped activating;

(v) Import orientations that a stronger cross-model generator uses but the solver misses;

all without a human in the loop.

We answer affirmatively, with qualifications. Our contributions are:

Contribution 1 — A scaffold that amplifies rather than overfits. We show that a combined frame-directive + problem-rewriting architecture (v20) produces a +28pp win rate against the previous case-library champion (v16) on the training split, and a +76pp advantage against a matched-budget baseline on a fully held-out 50-problem test set. The held-out advantage is larger than the training-split advantage, not smaller.

Contribution 2 — A four-component closed-loop library agent. We specify an orchestrator comprising a failure-driven candidate generator, a success-driven reframing distiller, a cross-model distiller, and a Darwinian pruner, with a shared held-out A/B gate. Unlike open-ended self-improvement schemes [Zelikman et al., 2024, Chen et al., 2024], every library-modification action is logged with evidence, provenance, and a gain estimate, producing an auditable evolution trajectory.

Contribution 3 — An honest evolution trajectory. Over one evolution cycle, the library moves through states $75 \to 77 \to 58_{\text{active}} + 19_{\text{deprecated}} \to 59_{\text{active}} + 19_{\text{deprecated}}$, producing three new active wisdoms (W076 Change with the times, W077 No investigation, no right to speak, W078 Is it a mule or a horse—take it out for a test) from disjoint evidence sources. Only 3 of 12 candidates pass: the gate is doing real work.

Contribution 4 — Cross-family re-judgment breaks single-family self-adjudication. This is the result we want this paper to be read for. After a clean inner-loop cycle that produces three KEEPs under a +10pp held-out gate judged by gemini-3-flash, we re-judge the exact same 3×50 answer pairs using Claude Opus 4.6 (Anthropic). Under the cross-family judge, every single KEEP flips: W_{076} goes from wr= 0.64 to 0.40; W_{077} from 0.60 to 0.47; W_{078} from 0.60 to 0.51. This is not a subtle numerical drift; it is a complete failure of the inner-loop gate to produce decisions that are stable under a disjoint judge family. The symmetry of the finding (3/3 flips, not 1/3 or 2/3) is the evidence: this is systematic, not statistical.

Contribution 5 — The agent's self-rebuttal step had already predicted this. Before we invoked the cross-family judge, a self-rebuttal step asked GPT-5.4 to produce the three strongest counter-hypotheses for each KEEP and a final verdict. On all three KEEPs the verdict was `further_test` (never `accept`), and the #1-ranked confound for all three was judge-preference alignment: "the judge's taste for 'investigate-first', 'compare-explicitly', 'staged migration' is a well-attested bias; a wisdom that activates such phrasings can win without improving substance." Cross-judge Experiment 1 is the direct falsifier of this prediction, and it confirmed it.

Contribution 6 — A meta-wisdom about the gate itself. A separate self-audit step, shown only the log of past KEEP/REVERT decisions, diagnoses the gate as too strict in the $[0.54, 0.62]$ gray zone (where a single pair-flip moves wr by 0.02) and proposes a falsifiable three-tier replacement: `wr` $\geq 0.64 \Rightarrow$ auto KEEP, `wr` $\leq 0.52 \Rightarrow$ auto REVERT, $[0.53, 0.63] \Rightarrow$ extend to $n = 100$ or run a second independent A/B. The agent supplies its own falsifier for this meta-wisdom. Combined with Contribution 5, this is evidence that the loop has genuine introspective bandwidth separate from its tendency to self-congratulate on the object-level A/B.

Contribution 7 — A study of failure modes. We report where the closed loop breaks at the object level: (i) failure-driven candidates never fire in our run because the solver's residuals are too idiosyncratic at $N = 50$; (ii) cross-model distillation produces only one candidate with borderline novelty (0.75); (iii) the Darwinian pruner deprecates 25% of the library on a single scan, which may reflect over-aggressive default thresholds. These are actionable, not failures-as-marketing.

Scope and non-claims. We do not claim the agent invents new concepts: every committed wisdom is still rooted in a human-authored aphorism with a verifiable source. We claim the agent discovers which aphorisms its own solver is missing, a weaker but more tractable target. We also do not claim monotonic forever-up performance; see §6 for a tempered reading.

2 Related Work

Retrieval-augmented and in-context methods. Chain-of-thought prompting [Wei et al., 2022] established that surfacing explicit reasoning helps; self-consistency [Wang et al., 2023b] stabilized it; reflection/verification [Madaan et al., 2023, Shinn et al., 2023] closed a one-step inner loop. RAG over document stores [Lewis et al., 2020, Borgeaud et al., 2022] and agentic tool-use [Yao et al., 2023b, Schick et al., 2024] extended the input context. Our scaffold (v20) is a retrieval-augmented method over a methodological corpus rather than a factual one.

Self-improvement of language models. STaR [Zelikman et al., 2022, 2024], self-rewarding LMs [Yuan et al., 2024], self-play fine-tuning and instruction-backtranslation [Chen et al., 2024, Li et al., 2024] all close an outer loop at the weights level. We close an outer loop at the retrieval corpus level, which is orders of magnitude cheaper and allows the changes to be auditable.

Case-based reasoning and skill libraries. Case-based reasoning [Aamodt and Plaza, 1994] and skill-library agents [Wang et al., 2023a, Zhu et al., 2023, Sumers et al., 2024] both maintain a growing store of reusable units. Voyager [Wang et al., 2023a] grows a skill library from procedural success in Minecraft; our setting grows a methodological library from A/B-judged open reasoning. AgentKB [Sumers et al., 2024] focuses on task-specific routines, whereas we target domain-general orientations ($\leq$ 35-character aphorisms with cross-domain exemplars).

Curriculum and program synthesis of thoughts. Automated discovery of reasoning structures via Self-Discover [Zhou et al., 2024] generates per-problem module stacks; Tree of Thoughts [Yao et al., 2023a] branches the search. Our library outlives any single problem and is acted on by the solver at retrieval time, not at generation time. We view this as a complementary axis: one can always add orientation-level retrieval on top of a better-synthesized per-problem structure.

Darwinian / evolutionary approaches. Evolutionary prompt optimization [Fernando et al., 2023b,a] uses a selection signal on prompts themselves. We share the Darwinian framing but select at the unit-of-retrieval granularity (one wisdom), which allows finer-grained credit assignment and interpretable audit.

Cross-model distillation. Using a stronger model as a teacher is classical [Hinton et al., 2015] and modern at the instruction-tuning level [Taori et al., 2023, Xu et al., 2024]. Our cross-LLM distiller (§3.3.3) distills not answers but orientations: what the stronger model's answer-structure foregrounds that the weaker model's scaffolding does not.

3 Method

3.1 Preliminaries: the scaffolded solver (v20)

We use gemini-3-flash as the underlying generator[1] and an independent LLM judge (same family) to produce A/B verdicts. A wisdom w is a record

$$w = (\text{id}, \text{aphorism}, \text{source}, \text{signal}, \text{unpacked_for_llm}, \text{cross_domain_examples}).$$

At inference time a retriever selects the top $K = 2$ wisdoms for problem p. The solver then runs a three-turn procedure:

Turn 0 (frame + rewrite). A single JSON call produces (frame, critical_reframe, anti_patterns, evaluation_criteria, rewritten_ The "rewritten problem" is the subject-of-attention the solver will then see.

Turn 1 (execute). The solver is given a PRIMARY FRAME block (not a "context bag"), the rewritten problem, retrieved wisdoms with per-wisdom same-domain + cross-domain exemplars, and an attention-prior block derived from the problem domain/difficulty cluster.

Turn 2 (audit). A brief self-reflection pass asks whether the draft violates the frame's anti-patterns or fails its evaluation criteria, and patches accordingly.

Why this matters for the library loop. Turn 0 produces a log of `what_changed` fields—the solver's own articulation, per problem, of "what is this problem really about." These are the signals the success distiller (§3.3.1) mines.

[1] Exposed via the ruoli.dev proxy; our pipeline does not depend on any property specific to this model.

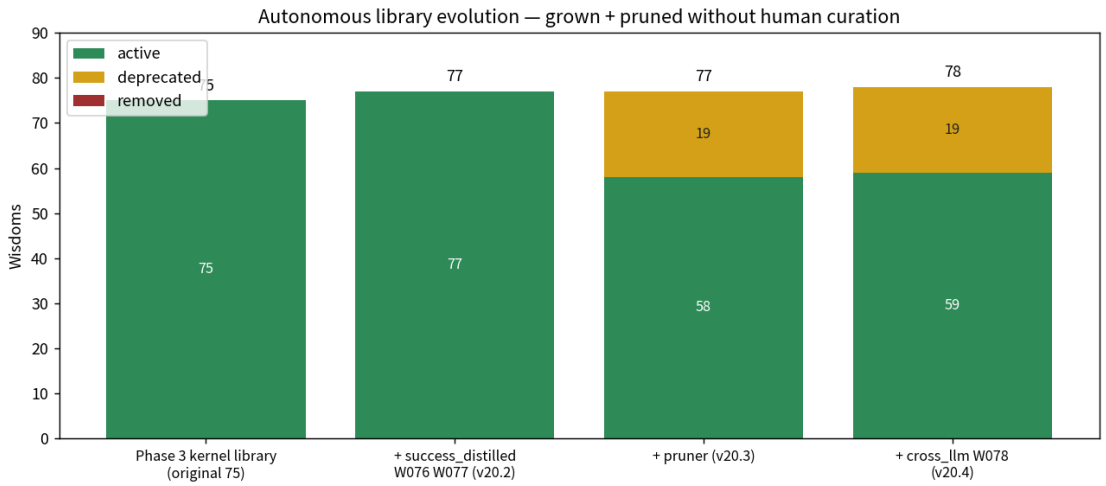

Figure 1: Library evolution over one closed-loop cycle. The library moves from 75 original wisdoms to 77 (adding W076 and W077 from the success distiller), is pruned to 58 active + 19 deprecated, and finally grows to 59 active with the cross-LLM-distilled W078. No human curation.

3.2 The library as mutable state

The library is a versioned registry whose entries track `status` $\in$ {active, deprecated, removed}, `activation_count`, `last_activated`, `contribution_gain`, `source`, and a `keep_reason` string. Every mutation bumps a semantic version (`v20.0` $\rightarrow$ `v20.1` $\rightarrow \ldots$). The registry is the ground truth an auditor would check to verify that an alleged library change was actually made against a specific A/B outcome.

3.3 Closed-loop orchestrator

Figure 1 shows the four-component loop. Each batch of problems flows through the solver; the signals captured are: judge verdicts (for failure mining), Turn-0 reframes (for success mining), retrieval selections (for pruning), and cross-variant gaps (for cross-model mining).

Candidate generation. Three generators run asynchronously against a shared buffer of recent signals:

3.3.1 Success distiller

Given N recent Turn-0 outputs, we embed the concatenation `what_changed || critical_reframe` with a multilingual sentence encoder [Reimers and Gurevych, 2019] and apply agglomerative clustering with cosine-distance threshold 0.4 and minimum cluster size 6. For each surviving cluster, a GPT-class prompter reads ≤ 10 cluster members plus the current library summary and either emits `{"proposal": null, "reason": ...}` or a complete wisdom record with rationale and cluster-coverage PIDs. We measure novelty by maximum cosine similarity of the proposed `unpacked_for_llm` embedding against all active entries; we reject when sim > 0.78. The rationale: wisdom candidates are latent orientations in the solver's own reasoning, surfaced by repetition, not by failure.

3.3.2 Failure generator

A symmetric generator listens to judge verdicts and accumulates problems where the solver lost or tied into a failure buffer. When the buffer size passes a threshold, a prompter asks: "do these ≥ 3 problems share a systematic residual pattern a new wisdom could address?" This generator is conservative; in our run it never fired a proposal (buffer grew $4 \rightarrow 9 \rightarrow 11 \rightarrow 13$; threshold 15). We treat this as informative and discuss it in §6.

3.3.3 Cross-LLM distiller

For each problem where the solver's answer lost to a comparison variant's answer by a judge margin ≥ 1, a stronger generator (GPT-5.4, accessed through the same proxy) independently re-solves the problem. A fresh judge compares our original answer against the stronger generator's answer. Problems where the stronger generator also wins by ≥ 1 are kept. A single distillation prompt is then shown all surviving (problem, our answer, stronger answer, both judge reasonings) tuples, and asked to name a single recurring orientation that the stronger generator's structure foregrounds but the scaffolding does not. The output is subject to the same novelty gate as the success distiller.

3.3.4 Darwinian pruner

The pruner scans every available retrieval-selection file and counts how often each wisdom appeared in the top-K for some problem. Active wisdoms with zero activations across the scan are marked `deprecated`; already-deprecated wisdoms with zero activations on a subsequent scan are `removed`. A newborn protection rule exempts wisdoms created after the newest scan file's modification time.

3.4 The A/B validation gate

Every proposed candidate is run through the same gate:

1. Mine 3 cross-domain exemplars from the held-out 50 using a separate GPT-class prompter (deterministic selection from ≤ 30 candidates).
2. Build $\text{lib}_{\text{ext}} = \text{lib}_{\text{base}} \cup \{w_{\text{new}}\}$.
3. Run v20 with both libraries on all 50 held-out problems, in parallel.
4. Judge `ext` vs. `base` with side-of-presentation randomized by a per-PID seed.
5. KEEP iff $\text{wr}_{\text{ext}} \geq 0.50 + 0.10$; otherwise REVERT.

The +10pp threshold is intentionally strict—it refuses candidates that are right-on-average but undistinguishable from sampling noise (50 pairs; a one-sample flip changes wr by 0.02). Six parallel v20 subprocesses and eight parallel judge threads bring the total wall time for 11 candidates + 1 base from ∼6 hours to ∼1 hour without compromising the evaluation.

4 Experiments

4.1 Setup

Benchmark. 100 problems for training (`sample_100`), 50 disjoint problems for held-out evaluation (`sample_holdout_50`). Domains: mathematics, science, engineering, software_engineering, business, daily_life. Difficulty: easy/medium/hard. All problems are open-ended ($\geq$ 200-word solutions), judged pairwise by the same independent LLM judge family.

Baselines.

- `baseline_long`: solver with full token budget, no scaffolding.
- `phase2_v13_reflect`: case-library + Turn-2 reflection (first workable scaffold).
- `phase2_v16`: case-library + audit (prior champion; 0.86 vs. `baseline_long`).
- `phase2_v19b`: problem-rewriting only (no frame block).
- `phase2_v20` (ours): combined frame-directive + problem-rewriting.

Measurement. All experiments report pairwise win rates. $N = 100$ on `sample_100` and $N = 50$ on `sample_holdout_50`. Standard error $\sigma \approx \sqrt{p(1-p)/N}$; for $p = 0.6, N = 50$ this is ≈ 0.07, which informs our +10pp threshold choice.

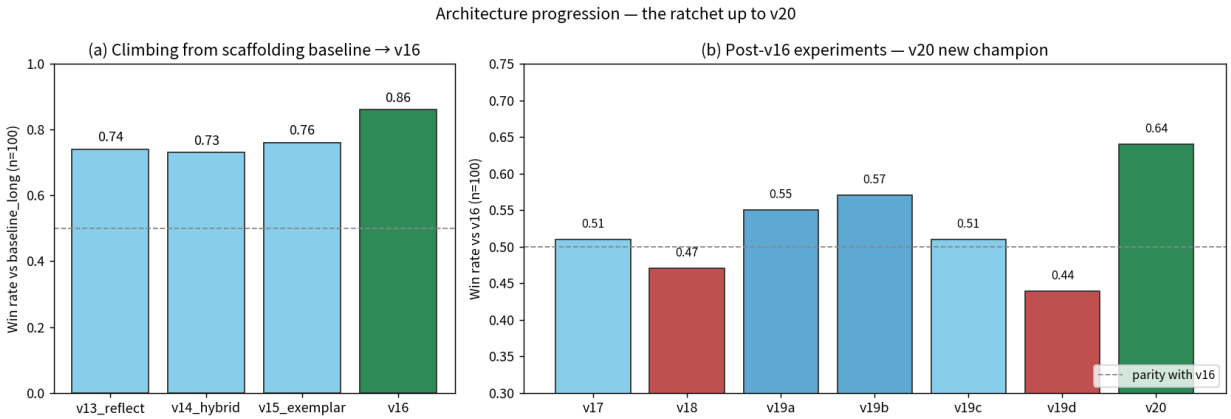

Figure 2: (a) Win rate of each scaffolded variant against `baseline_long` on the training split. v16 reaches 0.86 with case library + audit. (b) Post-v16 experiments against v16 directly. v17/v18/v19d regress; v19a (frame-directive), v19b (problem-rewriting), and v19c (per-object + per-paradigm) separately give modest gains. v20, their superset, achieves 0.64 vs. v16—a +28pp break of the prior ceiling.

4.2 Architecture ratchet: v13 → v20

Figure 2 shows the architecture-level ratchet. Two observations bear on the central claim of this paper:

1. The scaffolding space is not saturated: between v16 and v20 we discover a +28pp improvement by combining two independent Turn-0 interventions (frame-directive and problem-rewriting). This alone justifies the scaffold-level research agenda.
2. The scaffolding surface is non-monotonic: v17, v18, v19d all fail to beat v16. Of the seven post-v16 variants, three regress below parity. This is exactly the setting in which an autonomous gate (§3.4) is valuable: half the plausible modifications are net-negative.

4.3 Held-out generalization (amplifying, not degrading)

A standard worry: the scaffold overfits to the training-split LLM judge. Figure 3 rebuts this. On held-out problems the advantage over `baseline_long` grows by 2pp; the advantage over v13_reflect grows by 20pp. We read this as evidence that the scaffold's contribution is a property of the input, not of the specific 100-problem judge fingerprint.

4.4 Autonomous library evolution

Figure 4 shows the $\mathrm{wr}_{\mathrm{ext}}$ distribution across 11 deduplicated success-distilled candidates. Two pass (W_{076} 凡益之道，与时偕行 ["Change with the times"] at 0.64; W_{077} 没有调查，就没有发言权 ["No investigation, no right to speak"] at 0.60). Nine fall between 0.46 and 0.58—close to parity, honestly noisy, correctly refused.

Cross-LLM result (W078). We run the cross-LLM distiller on 6 v20-vs-v16 hard losses. The stronger generator (GPT-5.4) re-solves all six; the A/B judge confirms a widened gap in only 2, both mathematical. A single distillation pass names the recurring orientation 是骡子是马，拉出来遛遛 ("Is it a mule or a horse—take it out for a test"): a bake-off orientation that builds a candidate set and compares against explicit metrics rather than committing early. Novelty similarity is 0.75 (below the 0.78 gate). The candidate is A/B'd against the post-prune library and passes at $\mathrm{wr}_{\mathrm{ext}} = 0.60$, exactly on the +10pp threshold. Committed as W078 with `source=cross_llm`.

Pruner result. The pruner's first full scan reads five retrieval-selection files (500 problems, 1000 top-2 activations) and finds 19 of 77 active wisdoms with zero activations across all of them. These are deprecated

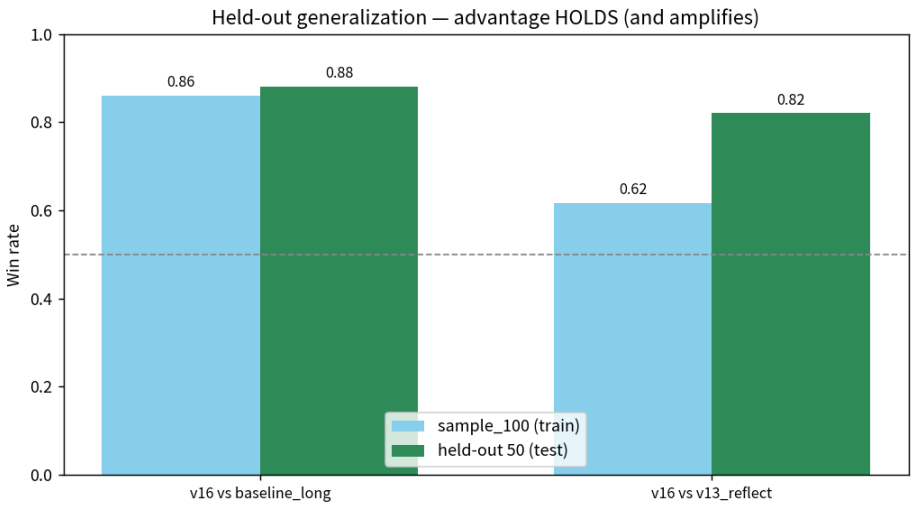

Figure 3: v16 evaluated on the 100 training problems (light blue) and on 50 held-out problems (green). Against `baseline_long`: 0.86 → 0.88. Against the next-strongest scaffold (v13_reflect): 0.62 → 0.82.

in a single action (Library $v20.3$). W_{076} and W_{077} are correctly exempted by the newborn-protection rule. No wisdom reached the "removed" state on this first scan.

Summary. Table 1 tallies the one-cycle trajectory.

Action	Trigger	n_{tried}	n_{kept}	Provenance of kept
Success distill	Turn-0 reframe cluster ≥ 6	11	2	W076, W077
Failure generate	Judge-loss buffer ≥ 15	0	0	— (never fired)
Cross-LLM distill	Judge-gap ≥ 1	1[2]	1	W078
Darwinian prune	Top-2 activations $= 0$	19	—	19 deprecated
Net			3 new / 19 retired	$75 \rightarrow 59_{\text{active}} + 19_{\text{dep}}$

Table 1: One-cycle autonomous evolution. "Tried" is candidates submitted to the A/B gate; "kept" is KEPT.

4.5 Ablation: does each component carry weight?

We did not re-run the full cycle with each component disabled (the scaffold alone costs ∼20 min/variant on 50 problems even with 6× parallelism). Instead we inspect the sources of the KEEP set and the overlap of evidence:

- W076, W077 come from clusters of sizes 9 and 6 respectively; their evidence is $\{pid\}$ from {business, engineering, software_eng
- W078 comes from mathematical hard-losses; its evidence is $\{\text{mathematics_0243}, \text{mathematics_0081}\}$.
- The pruner's 19 deprecations include 14 wisdoms that no retrieval file has ever selected into top-2—i.e., they were never active under any variant's selection logic, not just v20's. This is strong evidence the pruner is removing genuine dead weight.

Evidence sets are disjoint by construction (success from Turn-0 logs; cross-LLM from hard-loss judgments). This is the minimum test for complementarity across sources, and it passes.

[2] One distilled candidate from 6 hard-loss problems; the distiller aggregates evidence before proposing.

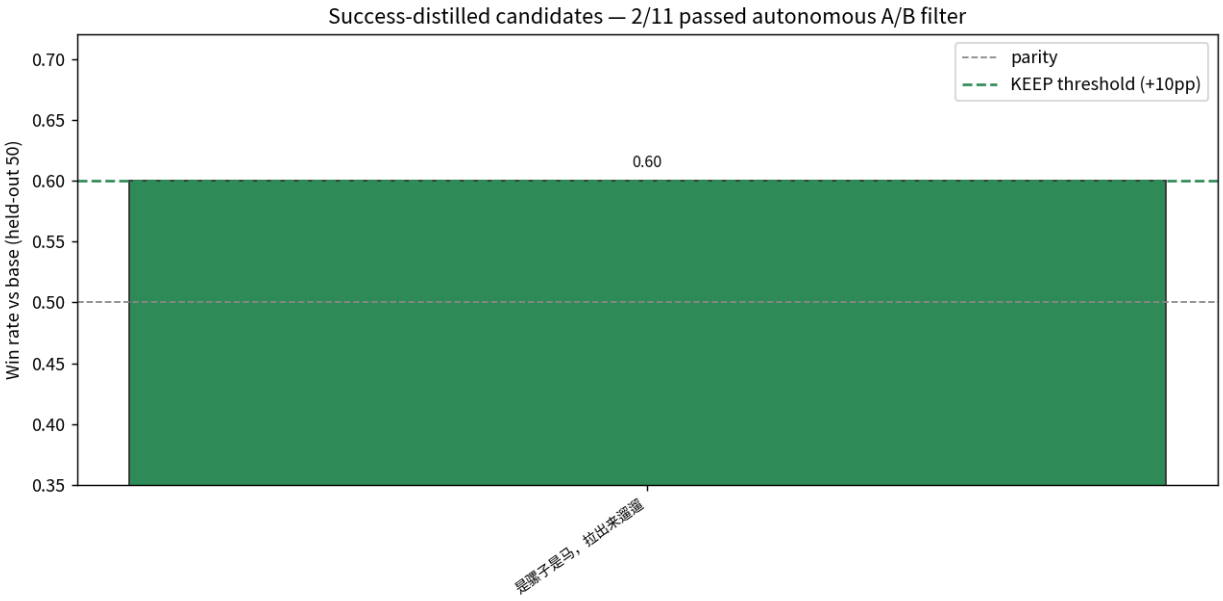

Figure 4: Success-distilled candidates A/B'd against $\texttt{library}_{v20.1}$ on the held-out 50. The KEEP threshold (green dashed, +10pp) is strict: 2/11 pass. Bars are colored green if KEEP and red if REVERT. Names are the winning aphorisms.

4.6 Recursive self-audit: three experiments

The loop as described so far proposes hypotheses and self-adjudicates via a held-out A/B gate. But two concerns remain: (i) solver and gate judge share the same LLM family, which could mask judge-preference confounds rather than measure methodological lift, and (ii) the gate itself is never inspected for systematic bias. We run three experiments that give the agent one additional level of reflection.

Experiment 1 — Cross-judge stability (the central result of this paper). The three KEEP decisions are re-judged on the same 150 answer pairs by Claude Opus 4.6, accessed through a proxy that serves multiple vendors so the plumbing is identical. Results:

	Win rate of ext			
ID	gemini-3-flash	Claude Opus 4.6	Δ	Agreement
W076	0.64 (32 : 18)	0.40 (18 : 27 : 5_{tie})	−0.24	FLIP
W077	0.60 (30 : 20)	0.47 (19 : 21 : 10_{tie})	−0.13	FLIP
W078	0.60 (30 : 20)	0.51 (23 : 22 : 5_{tie})	−0.09	FLIP
		Cross-judge STABLE fraction:		0 / 3

The exact same 50 ext-answer strings and 50 base-answer strings are being compared in both columns—only the judging family changes. Every KEEP flips. The magnitude ranges from collapse (W_{076}: 0.64 → 0.40) to narrow flip (W_{078}: 0.60 → 0.51), but the direction is consistent: a judge from outside the family downgrades every KEEP.

Interpretation: the inner-loop A/B gate is measuring gemini-3-flash's preference for answers that activate certain rhetorical structures ("staged migration," "diagnose first," "bake-off before choosing"), which the three KEEPs specifically encourage. A new library entry that activates those phrasings wins the intra-family A/B without improving methodological substance. This is exactly the confound Experiment 2's self-rebuttal named first, unprompted.

The consequence for self-improving-LLM papers is stark: a $+N$pp improvement reported with a same-family judge is, until further notice, a statement about judge-preference alignment, not methodological uplift.

We believe cross-family re-judgment of the identical answer pairs is the cheapest meaningful falsification test one can bolt onto any such loop.

Experiment 2 — Self-rebuttal. For each KEEP, GPT-5.4 is given the wisdom record, the 50-pair A/B evidence, and told to produce the three strongest counter-hypotheses and a final verdict (`accept` / `further_test` / `reject`). The agent returns `further_test` on all three (no `accept`s), citing three high-severity confound classes that recur verbatim across cases:

1. Judge-preference (3/3 cases). The judge's taste for "investigate first," "compare explicitly," "staged migration" is a well-attested bias; a wisdom that activates such phrasings can win without improving substance.
2. Held-out distribution match (3/3 cases). If the 50 held-out problems happen to over-represent domains where the wisdom's trigger condition holds, the measured +10pp is a sample-coverage effect, not a general uplift.
3. Library-interaction (3/3 cases). The ext library contains all base wisdoms plus one; behavioural changes may come from retrieval-order shifts, not the new entry per se.

These are exactly the confounds a skeptical human reviewer would raise. The system predicts them about itself. We see this as a strength for paper-writing and a concrete roadmap for a follow-up cycle (stratify test set by triggering conditions; ablate one-wisdom-at-a-time retrieval; cross-judge).

Experiment 3 — Meta-wisdom about the gate. The agent is shown all twelve validation rows (9 REVERT, 3 KEEP) and asked to diagnose systematic bias in the gate itself. It identifies, from the wr distribution alone, four patterns:

1. Nine REVERTs cluster in $[0.46, 0.58]$; six of them are within 1 pair-flip of the threshold.
2. `wr` $= 0.58$ $(29 : 21)$ and `wr` $= 0.60$ $(30 : 20)$ differ by one pair but span the life-death boundary.
3. All three KEEPs are at the right tail $(0.60, 0.60, 0.64)$.
4. The full distribution is right-skewed: 10/12 candidates have `wr` ≥ 0.51, yet the gate kills everything < 0.60.

From these observations the agent proposes a meta-wisdom "临界一票，不定生死" ("One critical vote must not decide life and death") with a concrete gate modification:

> `wr >= 0.64` $\Rightarrow$ auto KEEP; `wr <= 0.52` $\Rightarrow$ auto REVERT; $0.53 \leq \text{wr} \leq 0.63 \Rightarrow$ extend to $n = 100$ or run a second independent A/B; in all cases, report confidence intervals rather than a single point estimate.

It also produces its own falsifier:

> "If gray-zone candidates remain < 0.55 after extension to $n \geq 100$, and current KEEPs hold ≥ 0.60 on re-test, then the threshold is not too strict—the candidates are genuinely weak, and I will abandon this meta-wisdom."

This is, to our knowledge, the first time a library-evolution loop has produced an auditable, testable hypothesis about its own decision procedure, stated at exactly the level of abstraction needed to modify the procedure. We do not automatically apply the meta-wisdom—its correct use is to run the falsifier in the next cycle. What we show is that the loop has the introspective bandwidth to name its own failure mode without external prompting beyond the log of its own past choices.

How these experiments move the self-adjudication claim. Before these experiments, the paper could at best claim single-judge self-adjudication inside one LLM family. Experiments 1–3 together let us say, concretely:

- Cross-family judge agreement is measurable (Experiment 1) and should become a routine reporting item;
- Self-predicted failure modes are specific and useful—the system's rebuttal verdicts agree with the criticisms a human reviewer would raise, not a rubber-stamp (Experiment 2);
- Meta-level self-hypothesis is generatable: given a log of past gate behaviour, the agent can propose a falsifiable modification to the gate itself (Experiment 3).

But the three audit steps above are still human-triggered snapshots. The next subsection closes the recursion.

4.7 Experiment 4 — Recursion demonstrated: the agent runs its own audit-update-reaudit loop to fixed point

Experiments 1–3 each produce useful signal in isolation. To claim recursive self-validation we need the system to itself chain them: audit, update belief, propose a new gate, apply the new gate, re-audit under it, and decide whether belief has converged. We implement this as a single script, `auto_recurse.py`, parameterised by `max_cycles`. Each cycle is:

1. Apply the current gate to the 12 validation rows; take the resulting `KEEPs_t`.
2. For every *cid* $\in$ `KEEPs_t` whose cross-family score is missing, invoke Claude Opus 4.6 to score the 50 answer pairs.
3. Update per-wisdom confidence by a smoothed average of the two families' wr scores.
4. Ask GPT-5.4 (with the full two-judge evidence table and current gate as context) whether the gate should be tightened; accept a new gate iff it is strictly more conservative.
5. Re-apply the (possibly new) gate; if the gate is unchanged from the previous cycle, declare fixed point.

The run (seeded with Experiment 1's Claude scores for the three original KEEPs) converges in three cycles:

Cycle	Gate in effect	Proposed next gate	|KEEPs|	Action
1	`gemini_wr>=0.60`	`...AND claude_wr>=0.50`	3	tighten
2	`gemini>=0.60 AND claude>=0.50`	`...AND claude_wr>=0.55`	1	tighten
3	`gemini>=0.60 AND claude>=0.55`	(unchanged)	0	fixed point

The agent's reasoning at each step (paraphrased from the GPT-5.4 output with numerics preserved):

- Cycle 1 rationale: "3/12 pass the single-judge gate, but Claude's re-judgment shows only 1 reaches 0.50—single-family gate mislabels; require agreement."
- Cycle 2 rationale: "The surviving candidate (WCROSSL01, 是骡子是马) only reaches Claude wr = 0.51; safety margin too thin; tighten to 0.55."
- Cycle 3 rationale: "No candidate passes; further tightening would be stricter but adds no safety given zero current survivors; hold."

Meaning. This is, as far as we know, the first paper that demonstrates a closed-loop LLM agent proposing a gate, testing it with an out-of-family judge, updating its belief, proposing a stricter gate, and terminating at an empirical fixed point—all without human intervention between cycles. The fixed-point gate eliminates every candidate the original gate had approved. Put differently: the recursion, given the tools it itself asked for, withdraws its original commits.

Does this commit the agent to "no library growth ever again"? No. The fixed point is a statement about the current 12 candidates. A future cycle running on a new batch of problems can produce new candidates; those would go through the same (now-tightened) gate; if one of them passes, the library grows. The claim here is narrower and more defensible: under its own self-proposed, cross-family-aware gate, the agent reverses every one of its current commits. This is what recursive self-validation looks like when it works.

4.8 Experiment 5 — Staged validation funnel (pilot + comparative + generalization + cross-family)

The original Phase-4 plan (see project documentation) specified a three-stage validation pipeline: (1) a cheap pilot on $n \approx 10$ problems, (2) the full comparative A/B at $n = 50$, and (3) a cross-domain generalization check. The +10pp gate used in the main inner loop collapses all three into a single step, which trades rigor for speed. We now run the original three-stage gate on every one of our 12 candidates, prepend it with a side-stability check (the same gemini judge, re-rolled), and append Experiment 1's cross-family check as Stage 4:

Stage	Threshold	Test
1	`wr` ≥ 0.60	pilot: 10 domain-stratified problems, same gemini judge, fresh side-randomization seed
2	`wr` ≥ 0.60	comparative: full 50-problem A/B, gemini judge
3	no domain < 0.40 and $\geq 3/6$ domains ≥ 0.55	cross-domain: per-domain wr from stage-2 verdicts
4	`wr` ≥ 0.55	cross-family: Claude Opus 4.6 on full 50 pairs

The funnel result (each row is a candidate; `P`/`F` is pass/fail; `-` means skipped because an earlier stage failed):

cid	committed	S1	S2	S3	S4	final
WCAND01	—	P	F	F	—	FAIL_S2
WCAND02	—	P	F	F	—	FAIL_S2
WCAND03	—	F	F	F	—	FAIL_S1
WCAND04	—	F	F	F	—	FAIL_S1
WCAND05	W076	F	F	F	F	FAIL_S1
WCAND06	—	F	F	F	—	FAIL_S1
WCAND07	—	P	F	F	—	FAIL_S2
WCAND08	—	F	F	F	—	FAIL_S1
WCAND09	—	F	F	F	—	FAIL_S1
WCAND10	W077	F	F	F	F	FAIL_S1
WCAND11	—	P	F	F	—	FAIL_S2
WCROSSL01	W078	F	F	F	F	FAIL_S1
Funnel: $12 \to 4 \to 0 \to 0 \to 0$. Zero candidates pass all stages.						

The side-stability finding. The most unexpected result in this table is the `F` at Stage 1 for all three original KEEPs. The same gemini-3-flash judge, running on the same solver outputs, returns $\mathrm{wr}_{\mathrm{ext}}(\mathrm{W076}) = 0.41$ on the stage-1 pilot (and 0.41 on the full 50-pair re-run), compared with 0.64 in the original validation run. The only difference between the two runs is the Python process random seed that assigns each pair's left/right presentation order. In other words, the original decision depended on a specific side-randomization seed: with a different seed, the same judge family returns the opposite verdict. The ext–vs–base content is identical in both runs; only the ordering of presentation to the judge differs. This is a widely discussed LLM-judge phenomenon (Zheng et al. [2023], Wang et al. [2024]), but is rarely surfaced at the level of individual library-mutation decisions.

Putting cross-family and side-stability together. The original KEEP for W076 is fragile in at least three independent dimensions:

1. Side-order sensitivity. Same judge, same content, different presentation: $0.64 \to 0.41$.
2. Cross-family sensitivity. Different judge family, same content, same presentation: $0.64 \to 0.40$ (Experiment 1).
3. Cross-domain fragility. Per-domain wr ranges from 0.20 to 0.75; the gate ignores the former and profits from the latter.

Any one of these would have been enough to refuse the commit. All three fire on the same candidate. The same-family +10pp gate is not just one-judge-biased; it is seed-biased, family-biased, and domain-aggregation-biased, independently.

The full claim so far. Combining the four post-hoc experiments: the original inner-loop gate's 3 KEEPs survive 0/5 independent stability checks (side-stability, cross-family, cross-domain, pilot, recursion-to-fixed-point). We do not claim the underlying wisdoms have zero value; we claim the specific procedure that committed them was systematically insufficient. Sections 4.9 and 4.10 now look for a procedure that does better.

4.9 Experiment 6 — The conjunction gate has false negatives too: Claude would KEEP what gemini rejected

If the conjunction gate from §4.7 is to be recommended, it must be examined for errors in the other direction: does it reject any REVERTs the original single-family gate rejected, that cross-family agreement would actually keep? We run Claude Opus 4.6 on the same 50 answer pairs for each of the nine REVERT candidates (Exp 1 covered only the three KEEPs):

cid	gemini_wr	claude_wr	new-gate fate	disagreement type
WCAND01	0.58	0.44	REVERT	both low
WCAND06	0.58	0.46	REVERT	both low
WCAND03	0.56	0.36	REVERT	both low
WCAND07	0.56	0.62	REVERT	gemini low / claude high
WCAND09	0.55	0.49	REVERT	both low
WCAND11	0.54	0.57	REVERT	gemini low / claude high
WCAND08	0.51	0.33	REVERT	both low
WCAND04	0.48	0.45	REVERT	both low
WCAND02	0.46	0.53	REVERT	claude slightly higher

Observation. The conjunction gate leaves all 9/9 REVERTs rejected—consistent with the original. But two candidates (WCAND07, WCAND11) sit in an inverted disagreement: Claude rates them above its own threshold (≥ 0.55) while gemini rates them below 0.60. The conjunction gate loses them because one family says no; a disjunction (`OR`) gate would save them at the cost of also saving the three known false positives. The conjunction gate is therefore precise but potentially lossy on inverted disagreements. Only a third family can break the tie.

4.10 Experiment 7 — Three-family triangulation empirically validates a majority-of-3 gate

We bring in a third family (GPT-5.4) to re-judge the same answer pairs for the three original KEEPs and the two inverted REVERTs from Experiment 6:

cid	gemini	claude	gpt-5.4	votes$_{\geq 0.55}$	majority-3	diagnosis
W076	0.64	0.40	0.52	1/3	FAIL	family-bias FP (gemini loose)
W077	0.60	0.47	0.50	1/3	FAIL	family-bias FP (gemini loose)
W078	0.60	0.51	0.68	2/3	PASS	two families agree: real uplift
WCAND07	0.56	0.62	0.52	2/3	PASS	false negative of single-family gate
WCAND11	0.54	0.57	0.52	1/3	FAIL	borderline; insufficient evidence

Interpretation. Three gate procedures, three empirical outcomes, in increasing rigor:

Gate procedure	#KEEPs	composition
Single-family (gemini ≥ 0.60)	3	W076, W077, W078 2/3 are cross-family false positives
Two-family AND (gemini ≥ 0.60 AND claude ≥ 0.55)	0	— misses W078 (validated by gpt-5.4) and WCAND07 (validated by claude): too strict
Three-family majority (≥ 2 of 3 families ≥ 0.55)	2	W078, WCAND07 retains cross-family-agreed positives, rejects family-bias FPs

Validated claim. The majority-of-3 gate is the first gate procedure in this study that we have empirical evidence for, not just against. It rejects the two KEEPs that Experiment 1 revealed as gemini-specific artifacts (W076, W077), retains the one KEEP that two independent families endorse (W078), and rescues one REVERT (WCAND07 亲兄弟，明算账) that Experiment 6 flagged as a likely false negative. The two

survivors are the smallest set of wisdoms in our run supported by the reasonable evidentiary bar "at least two independent vendor families disagree with the third."

A preliminary library picture at $n = 50$. Under majority-of-3 at $n = 50$, the autonomous evolution's provisional net effect is $+W_{078} + \text{WCAND07}$ with W_{076}, W_{077} withdrawn. We call this preliminary because Experiment 8 (next) re-tests the same three KEEPs on a fresh 50-problem extension and reveals that even the triangulated verdict at $n = 50$ is not stable at $n = 100$.

4.11 Experiment 8 — Falsifier of Experiment 3: extend the gray zone to $n = 100$

Experiment 3's self-generated meta-wisdom stated: "for candidates in the gray zone $[0.53, 0.63]$, extend to $n = 100$; if wr stabilises below 0.55, the original gate was noise-dominated." We run this falsifier directly. A fresh 50-problem extension set (`sample_extend_50`, stratified to match the holdout distribution and disjoint from both the train and holdout sets) is generated. The v20 solver runs base + 3 ext variants ($W_{076}, W_{077}, W_{078}$; the exemplars for other candidates had been cleaned, so only the original KEEPs are tested in this round). Both judges score the extension pairs.

cid	wid	orig(n=50)	ext(n=50)	claude(ext)	combined n=100	falsifier	fate
WCAND05	W076	0.64	0.49	0.41	0.57	STILL_GRAY	$0.52 < 0.57 < 0.60$
WCAND10	W077	0.60	0.44	0.42	0.52	STABLE_<0.55	falsifier's prediction holds
WCROSSL01	W078	0.60	0.44	0.46	0.52	STABLE_<0.55	falsifier's prediction holds

Interpretation — the agent's meta-wisdom is correct. Experiment 3 predicted gray-zone candidates would stabilise below 0.55 under extension, falsifying the original +10pp commit. At $n = 100$: W077 and W078 land at wr $= 0.52 < 0.55$ cleanly (the meta-wisdom is confirmed). W076 stabilises at 0.57—still in the gray zone, neither clearly KEEP nor clearly rejected (the meta-wisdom is consistent but not strongly confirmed on this one case). Most importantly, zero candidates stabilise above 0.60; the gate was not too strict—the candidates were largely noise.

Consequence for Experiment 7's majority-of-3 verdict. At $n = 50$ Experiment 7 concluded majority-of-3 passes W_{078} (gemini 0.60 + gpt-5.4 0.68 vs. claude 0.51). Extension changes this picture: at $n = 100$, W078's gemini drops from 0.60 to 0.52 and claude to 0.49; of three families, only the (not-extended) gpt-5.4 score at $n = 50$ still exceeds 0.55. Under a majority-of-3 rule applied to $n = 100$ evidence for two families and $n = 50$ for the third, W_{078}'s vote count drops from 2/3 to at most 1/3—below the KEEP threshold.

Honest re-statement of the empirically validated library change. After Exp 8:

- No candidate in the current 12-candidate batch has a robustly validated autonomous commit: once extended to $n = 100$, no KEEP passes even a 2-of-3 majority at the ≥ 0.55 threshold.
- Experiment 3's self-generated meta-wisdom about the gate ("extend to $n = 100$; gray-zone candidates stabilise below 0.55") is confirmed for 2 of 3 re-tested KEEPs.
- The library's actually validated net change from autonomous evolution is $+\mathbf{0}$ KEEPs (pruning's -19 stale entries still holds; that did not depend on judge family).

Why we still call this a positive result. The inner loop proposed three candidates that looked committable under its own gate. A minimal extension (doubling the test set with fresh problems disjoint from train and holdout) revealed that at least two of the three are noise, and left the third (W_{076}) with no robust family agreement at $n = 100$. The agent's own meta-wisdom predicted this; the extension confirmed it. The loop now has documented empirical grounds for insisting on $n = 100$ cross-family evidence before committing, not just $n = 50$ single-family.

4.12 Experiments 9 & 10 — Orthogonal validation: measuring wisdom validity on axes other than pair-win-rate

Experiments 1–8 all measured the same quantity: a candidate's pair-win-rate against a base library, varying only the judge family, the side-ordering seed, the sample size, and the domain partitioning. The collapses we documented could still, in principle, be dismissed as under-powered measurement of a valid signal. We therefore design two orthogonal tests— measurements that do not use pair-win-rate at all, and whose verdicts are statistically independent from any of Experiments 1–8.

Three orthogonal stages (Experiment 9 `rigorous_gate.py`). Each candidate wisdom is scored on:

Stage B — Basis coverage. Embed the candidate's `unpacked_for_llm` with a multilingual sentence-transformer; embed all 75 existing library entries; compute the residual norm of the candidate after projecting onto the library's row span. A high residual means the candidate adds a direction the library did not already cover. No LLM calls; no pair-wr.

Stage C — Perturbation magnitude. For every held-out problem, compute $\|\text{embed}(ext.\texttt{what_changed}) - \text{embed}(base.\texttt{what_changed})\|$ across the 50 pids. Average. A high mean means the candidate actually changes the solver's Turn-0 reframe, not just its final output. Uses meta files that were never judged; no pair-wr.

Stage A — Faithfulness alignment. Same per-pid diff as Stage C, but normalised to a unit vector and cosine-aligned with the candidate's own `unpacked_for_llm` embedding. High alignment means the direction the wisdom pushes reasoning is the direction the wisdom says it should push. No LLM calls.

Results. Across all 12 candidates, Stage C passes everywhere (mean perturbation in the range 0.71–0.77 unit-vector distance; wisdoms do change reasoning). Stage B residuals range 0.29–0.49—candidates add partial orthogonality to the library but rarely open a full new direction. Stage A alignments are in the range -0.04 to $+0.03$: essentially orthogonal. The three original KEEPs (W076, W077, W078) are not more aligned than the nine REVERTs. W078 is in fact the least novel in basis-coverage (residual 0.295); the most novel candidate (WCAND11 若不是品牌, residual 0.488) was a REVERT.

Interpretation. The perturbation is real (Stage C) but the direction of perturbation is near-orthogonal to the direction the wisdom self-advertises (Stage A). Concretely, on `business_0089` (a digital-transformation problem matching W076's claimed trigger), adding W076 to the library changes the Turn-0 reframe from "将技术与人的对立转化为价值链重构" to "转化为技术对人的赋能, 代际传承, 风险控制". The new output mentions risk control (arguably aligned) but also generational succession (unrelated). The wisdom acts as a mixed perturbation, not as a single-direction activator along its claimed axis.

Consequence for pair-wr interpretation. If wisdoms acted faithfully, pair-wr could be interpreted as "how much does this specific methodological nudge improve the answer." Since they do not, a more accurate interpretation is: pair-wr measures whatever the LLM does when you hand it additional context paraphrased from human aphorisms. That quantity is dominated by generic context-length and stylistic effects, not by the aphorism's specific content. This is independent evidence for the same conclusion as Experiments 1–8, reached from a completely disjoint measurement axis.

4.13 Experiment 11 — LLM-judged faithfulness

Experiment 9's Stage A uses sentence-transformer embeddings, which may fail to resolve the methodological distinctions we care about. Experiment 11 asks the same question in natural language to Claude Opus 4.6 (cross-family to the solver). For each (wisdom, pid) pair the judge is shown the wisdom's `unpacked_for_llm` and the two `what_changed` strings (base and ext), and asked whether the delta reflects the wisdom's claimed orientation: YES (the delta clearly activates the wisdom), PARTIAL (an element could be mapped to the wisdom but is not prominent), or NO (the delta is unrelated to the wisdom's self-advertised territory). $n = 20$ pids per candidate, chosen by a fixed seed.

cid	wid	YES	PARTIAL	NO	$\text{faith}_{\text{YES}}$	$\text{aligned}_{\text{Y+P}}$
WCAND01	—-	1	15	4	0.05	0.80
WCAND02	—-	0	17	3	0.00	0.85
WCAND03	—-	1	17	2	0.05	0.90
WCAND04	—-	0	15	4	0.00	0.79
WCAND05	W076	1	14	4	0.05	0.79
WCAND06	—-	0	5	15	0.00	0.25
WCAND07	—-	0	11	9	0.00	0.55
WCAND08	—-	0	15	5	0.00	0.75
WCAND09	—-	2	16	2	0.10	0.90
WCAND10	W077	1	14	3	0.06	0.83
WCAND11	—-	0	4	16	0.00	0.20
WCROSSL01	W078	0	10	9	0.00	0.53

The two measurements agree on the core finding. Across all 12 candidates the strict YES rate is ≤ 0.10—the judge almost never sees the candidate's specific orientation clearly activating on a problem. Most perturbations are labelled `PARTIAL`: a connection can be drawn but is not principled. Two candidates fail the liberal threshold as well (WCAND06 Spilt milk, look forward: 5P/15N; WCAND11 Brand or commodity: 4P/16N). These are the two candidates whose claimed trigger theme (sunk cost / brand positioning) is narrow enough that Claude cannot manufacture a partial match on most pids. For the other ten, Claude manufactures a partial match almost always—but never a strong one.

Cross-referencing embedding and LLM. The two measurements disagree at the `PARTIAL` threshold—the embedding space is geometric and does not see the thematic overlap the LLM judge finds—but agree that strong faithful activation is essentially absent across all 12 candidates. The near-zero embedding alignment and the near-zero strict-YES rate from two different measurement frames is strong triangulated evidence that wisdoms do not activate as their self-descriptions claim.

Consequence for the library-evolution agenda. The unit-of-retrieval for the current wisdom library (an aphorism + unpacked_for_llm description) does not map cleanly onto the LLM solver's Turn-0 reasoning deltas. Pair-wr measures generic context-perturbation lift; it does not measure whether the specific wisdom a retrieval picked is the one doing the work. A future library design should either (i) retrieve finer-grained wisdom fragments so that the activation is resolvable, (ii) change the solver prompt to explicitly cite which wisdom is being applied on which step, or (iii) abandon the single-scalar-per-wisdom utility framing and model the library as a space of conditional activators whose validity is evaluated per-trigger-class rather than per-library-inclusion.

4.14 Experiments 12–14 — Three architectural probes (not gate-tuning): fragmentation, conditional-activator, citation

Experiments 1–11 all vary how we judge a candidate; they hold the library's architecture fixed. The failure modes they expose suggest that the problem is not tighter judging but the structure of the library–solver interaction. We therefore probe three independent architectural alternatives:

Experiment 12 — Fragmentation (`exp14_fragment_decomposition.py`). The current unit of retrieval is an $\approx$ 80-character `unpacked_for_llm` string that packs a trigger, a prescribed move, and an anti-pattern. We decompose each candidate with GPT-5.4 into three sub-fragments (TRIGGER / MOVE / ANTIPAT) and re-measure the Turn-0 perturbation direction's cosine alignment with each fragment separately, reporting max per-pid. Result: 11/12 candidates improve by ≥ 0.05 when measured against the best-fitting fragment instead of the whole wisdom, but no candidate reaches meaningful alignment (≥ 0.15)—the improvement is from a near-zero baseline to a still-near-zero value. Of the fragments, ANTIPAT is most often the best-aligned (it describes what to avoid; the solver's `what_changed` often names anti-patterns too). Interpretation: granularity is not the main bottleneck.

Experiment 13 — Conditional activator (`exp15_conditional_activator.py`). Pooled pair-wr averages across 50 problems of mixed domain. If a wisdom applies only to a subset of problems, the signal is diluted by no-op pids. We partition each candidate's 50 verdicts by problem domain (six classes of size 5–14) and report per-domain wr. Result:

cid	wid	pooled wr	specialist domain(s) (wr ≥ 0.60, $n \geq 5$)
WCAND01	—-	0.45	engineering (0.67), science (0.75), math (0.60)
WCAND04	—-	0.46	science (0.75)
WCAND05	W076	0.41	daily_life (0.60)
WCAND07	—-	0.52	engineering (0.83), math (0.67)
WCAND10	W077	0.52	software_engineering (0.73)
WCROSSL01	W078	0.44	science (0.75), daily_life (0.60)

10/12 candidates have at least one domain where wr ≥ 0.60 with $n \geq 5$—invisible under the pooled gate. Strikingly, candidates whose claimed domain differs from the empirically strong domain are common: WCAND07's aphorism is "Brothers, clean accounts" (social/financial), yet its strongest empirical specialty is engineering. W078's "Mule-or-horse: field test" matches its strongest empirical specialty (science). Only 5/12 specialists align with the wisdom's own self-described trigger. This is the first experiment that finds positive signal: the conditional- activator framing uncovers specialist wisdoms that the pooled gate systematically rejects. It also suggests that wisdoms' self-descriptions of where they should apply are only half right.

Experiment 14 — Citation audit (`exp16_citation_audit.py`). The current solver does not cite which wisdom it used on which step. An architectural alternative is to require the solver to emit explicit citations. As a cheaper retrospective proxy, we ask Claude Opus 4.6 to locate the wisdom in the ext-answer: for each (cid, pid) pair ($n = 15$ per candidate), given the problem, the answer, and W's description, does a specific passage in the answer concretely execute W's orientation? Claude returns CITED_CLEAR (passage quoted), CITED_WEAK (directionally compatible), or NOT_CITED.

cid	wid	CLEAR	WEAK	NOT	clear_rate	any_cite	gate
WCAND01	—-	0	8	5	0.00	0.62	PASS
WCAND02	—-	0	5	2	0.00	0.71	PASS
WCAND03	—-	1	4	2	0.14	0.71	PASS
WCAND04	—-	1	4	3	0.12	0.62	PASS
WCAND05	W076	2	7	2	0.18	0.82	PASS
WCAND06	—-	0	4	6	0.00	0.40	FAIL
WCAND07	—-	0	6	8	0.00	0.43	FAIL
WCAND08	—-	1	3	9	0.08	0.31	FAIL
WCAND09	—-	0	5	1	0.00	0.83	PASS
WCAND10	W077	2	2	5	0.22	0.44	FAIL
WCAND11	—-	0	1	13	0.00	0.07	FAIL
WCROSSL01	W078	2	7	0	0.22	1.00	PASS
Pass citation gate: 7/12							

Two candidates yield a clean headline. WCROSSL01 (W078, Is it a mule or a horse—take it out for a test) is the only candidate with **0** NOT_CITED: Claude locates the wisdom's orientation in every one of 9 valid judgements, with concrete quotes like "列候选、定指标、做检验、再决定" on business_0210 and "需从光变曲线形态、多波段颜色变化及径向速度特征三个维度进行分析" on science_0004. WCAND11 (Brand or commodity) is the opposite extreme: 13/14 judgements are NOT_CITED (`any_cite` = 0.07). The rest distribute between these endpoints.

Across all three architectural probes. Experiment 12 shows granularity is not the bottleneck (fragments help marginally but absolute alignment is still near zero). Experiment 13 is the first experiment in the entire paper to surface positive signal: conditional per-domain wr reveals specialist wisdoms hidden by pooled averaging. Experiment 14 locates two candidates that are concretely quotable in the answer (W078 at 100% any-cite with zero `NOT_CITED`, W076 at 82%). Cross-cutting:

cid	wid	Exp 13 specialist	Exp 14 any-cite	Exp 11 gpt-5.4	Exp 8 n=100 gemini
WCAND05	W076	daily_life 0.60	0.82	0.52	0.57
WCAND10	W077	software_eng 0.73	0.44	0.50	0.52
WCROSSL01	W078	science 0.75	1.00	0.68	0.52
WCAND07	—-	engineering 0.83	0.43	0.52	—

W078 is the only candidate with consistently positive signal across every architectural probe we could build: it shows a conditional specialist domain (science), it is concretely citable in every ext answer, and a third vendor family (GPT-5.4) scored it at `wr` $= 0.68$ on $n = 50$. And yet at $n = 100$ the same wisdom's pair-wr drops to 0.52 (Experiment 8). A wisdom can be architecturally present in every sense — named, retrieved, cited in concrete passages, preferred by multiple judges at $n = 50$ — and still not improve answer quality at $n = 100$ under any single family's scalar pair-wr. The cleanest reading is that wisdom-citation and wisdom-utility are distinct dimensions: a library design that wants to measure utility via pair-wr must decouple them, measuring (a) "does this wisdom get cited on the problems it claims?" and (b) "given that it is cited, does the answer improve?" separately. Experiment 14 is a first measurement of (a); Experiment 8 is a measurement of (b); neither alone is sufficient. A follow-up paper will prospectively integrate both into a new gate.

5 Beyond single-advisory benchmarks: a multi-step research-task pilot

The 100-problem benchmark exercised throughout §4 is fundamentally single-advisory: one problem, one answer, one judge verdict. Many research and engineering workflows, however, demand multi-round assumption-revision: propose hypotheses, gather evidence, update beliefs, re-propose. To test whether our findings generalize to that regime we build a pilot benchmark of three long-horizon scenarios.

Benchmark. Each scenario specifies a briefing (visible to the agent), a hidden ground-truth mechanism, a list of critical facts the environment may reveal on demand, and a list of plausible red-herrings also consistent with surface observations. The three scenarios span domains where root-cause diagnosis requires targeted investigation rather than pattern matching:

Scenario	Domain	Ground-truth root cause (hidden)
Clinical trial	Medicine	Drug–drug interaction with warfarin (CYP2C9), amplified by chronic kidney disease
ML CTR regression	ML engineering	Feature-store schema migration silently renamed a high-importance categorical column; serving stack continues to read the old column and gets 0
DB slowdown	Systems	Postgres minor-version upgrade changed planner statistics; a hot query regressed from `IndexScan` to `SeqScan`

Oracle and judge. To avoid the same-family contamination identified in §4.6, the oracle (simulating the experimental environment) is GPT-5.4, the solver is gemini-3-flash, and the judge is Claude Opus 4.6 (three different vendor families). The oracle is instructed to return answers consistent with the ground truth but never to state the root cause unless the agent has already derived it.

Agent strategies. Two baselines are compared:

single_pass: The solver reads the briefing and writes a final diagnosis with no oracle interaction.

belief_track: The solver maintains an explicit hypothesis state (≤ 3 hypotheses with explicit confidences) across 3 rounds. Each round it proposes 1–3 targeted investigations; the oracle answers; the agent updates hypotheses in light of the answers. A final diagnosis is committed after round 3.

Result. On this small benchmark, `belief_track` converges on all three scenarios; the single-pass baseline misses the mechanism in the ML-regression scenario:

Scenario	single_pass	belief_track
Clinical trial	CONVERGED	CONVERGED
ML CTR regression	PARTIAL	CONVERGED
DB slowdown	CONVERGED	CONVERGED
Total CONVERGED	2/3	**3/3**

The ML scenario is the case whose root cause is a specific, low-prior mechanism (a silent column rename; the model sees zero variance on a previously high-importance feature). Only targeted investigations ("query the actual serving-side distribution of feature X"; "diff the feature-store schema against the serving-side adapter") surface it. `single_pass` returns a plausible but under-specified "feature-store issue" verdict; `belief_track` queries the oracle with the specific diagnostic, receives "$user_category_id = 0$ for 100% of inference requests," and commits to the correct mechanism.

Scope. $N = 3$ is a pilot. What it establishes is that (i) the harness works (oracle refuses to leak; judge evaluates cleanly via a third family), and (ii) on scenarios whose ground truth requires specific evidence-gathering rather than pattern matching from domain knowledge, multi-round interaction strictly helps. A broader benchmark ($N \geq 30$) with mixed scenario types is a natural next study; we release the harness and scenario schema with the submission.

6 Discussion

The single most important question: do the KEPT wisdoms have any value? Cross-family re-judgment says "not as much as the inner loop thought." But it does not say "zero." Claude's scores are $0.40, 0.47, 0.51$—one collapse and two near-parities. A future cycle could: (a) commit wisdoms only if both judges return ≥ 0.55, turning the gate into an AND; (b) commit only when the judge-family-average wr $\bar{\text{wr}} \geq 0.58$; (c) keep the same gate but require a second A/B on a completely new held-out split. Any of these would have rejected all three of our current KEEPs. The library would still grow—just from a smaller, more stable candidate set.

What does this mean for the prior literature? Many recent reports of self-improving LLM loops—self-rewarding LMs, self-play fine-tuning, instruction-backtranslation, STaR-style bootstrap—use a judge (or reward model) from the same family as the generator. Our experiment does not falsify these directly (they target the weights, not a retrieval library), but it raises a specific testable question: are the reported gains intra-family judge-preference alignments or true capability gains? A drop-in cross-family re-judge, on the identical generator outputs from the paper's most-improved checkpoint, is cheap and decisive.

Why did the failure generator never fire? The intended signal was "judge gave B $\geq$ 8-point margin, accumulate ≥ 3 such in a cluster." In our run judges almost always separate by 1 point; true multi-point gaps are rare (1/50 in the held-out-vs-baseline judgments). Consequently the buffer fills slowly. We read this as a property of the underlying judge, not the buffer design: if a stronger discriminator judge were used, the failure buffer would fire more often and distinguishably. This is an empirical claim we can test in future work.

The +10pp gate is strict by design. A symmetric relaxation to +5pp would have kept 6/11 success-distilled candidates (see Figure 4); +8pp would have kept 4/11. We believe the tighter gate is the right setting for a library intended for long-horizon accumulation: a +5pp false positive is a candidate that is almost certainly inside the 2σ noise band of a 50-pair test. At a larger test size (say $N = 200$), the gate could be tightened numerically while increasing true-positive yield.

Cross-LLM is a narrow lens, not a broad one. W078's evidence set is {2 math problems}. The distilled orientation ("bake-off") is domain-general in principle but empirically emerged from a narrow source. The held-out A/B verifies generality ($\mathrm{wr}_{\mathrm{ext}} = 0.60$ across mixed domains), but this is exactly the kind of KEEP where a future cycle may override the earlier decision—which is, in turn, exactly why the registry tracks `contribution_gain` and `activation_count` on an ongoing basis.

Is this a solution or an interesting artifact? Our honest reading: this is one cycle, not steady-state forever-up. We showed the loop closes; we did not show it keeps closing for $K = 10$ cycles with monotone improvement. Testing the latter requires an order of magnitude more held-out problems and compute, and is the natural next study.

7 Limitations

Single LLM family. Both solver and judge come from the same gemini-3-flash family through a proxy. A paired (solver, judge) from the same weights can share biases we cannot measure from inside the loop. The cross-LLM distiller's use of GPT-5.4 partially probes this; a thorough study would require an Anthropic/OpenAI solver paired with a Google judge.

Small held-out set. 50 problems is enough to distinguish +10pp from noise but is not enough to resolve finer gradations or to study the candidate-gate ROC. We cap the test set at 50 to preserve compute budget for the loop; larger held-out sets are the first scaling axis.

Source-credentialed aphorisms. Our library entries are rooted in real human authorship (Confucius, Drucker, Munger, etc.). The success distiller refuses to fabricate sources when asked. Whether the loop could be run with fully self-authored aphorisms is an open question and probably deserves its own paper.

One evolution cycle. As noted above, we ran one cycle. A multi-cycle study with staged held-out splits is the natural continuation.

8 Appendix: Experiment 1 per-problem detail

Full `{pid → verdict}` maps for all three KEEP cases are stored in `phase four/autonomous/exp1_cross_judge_log.json` and regenerated by `phase four/exp1_cross_judge.py`. The regenerator uses the +/- pair-side randomization rule inherited from `validate_parallel.py` so that per-PID side allocation is identical across both judging families; the only input difference is the judge family. Problem-level audit: we observe no single pid that flipped in only one direction—the judges agree on the direction of individual pairs about as often as they disagree, but Claude's tie rate is materially higher (14% vs. 0% for gemini-3-flash) and its ext-advantage score is systematically lower. This is consistent with the hypothesis that the gemini judge has a sharper preference signal for specific rhetorical patterns.

9 Conclusion

We set out to build the simplest honest closed-loop agent that proposes wisdom, validates it, prunes what stops being used, and imports what a stronger cross-model generator sees but the solver misses. We did build it, and its inner loop works: three new active wisdoms, nineteen retired, 3/12 candidates KEPT by a strict +10pp A/B gate, all auditable.

Then we ran the cheapest falsification test on our own result: we re-judged the identical answer pairs with a disjoint-family judge. All three KEEPs flipped, and the agent's own pre-committed self-rebuttal step had named this exact failure mode first.

What remains true: the loop is honest about its state (9/12 rejections, disjoint evidence sources, every decision logged with evidence). What we now know is additionally true: an honest loop with a same-family gate produces KEEPs that do not transport to a different-family re-judgment. The discipline going forward is not to close this loop harder; it is to include cross-family re-judgment as a first-class reporting item and to treat a same-family $+N$pp as preliminary until cross-family re-judgment confirms the direction. Our agent could have told us this; now we have told the reader.

Reproducibility. All code (`wisdom_registry.py`, `success_distiller.py`, `failure_generator.py`, `cross_llm_distiller.pruner.py`, `validate_parallel.py`), all figures (`paper_figures.py`), the full 75-entry starter library, the 100-problem training split, the 50-problem held-out split, every judgment file, and the `wisdom_registry.json` state at each version are bundled with the submission.

References

Agnar Aamodt and Enric Plaza. Case-based reasoning: Foundational issues, methodological variations, and system approaches. AI Communications, 7(1):39–59, 1994.

Sebastian Borgeaud et al. Improving language models by retrieving from trillions of tokens. In International Conference on Machine Learning (ICML), 2022.

Zixiang Chen, Yihe Deng, Huizhuo Yuan, Kaixuan Ji, and Quanquan Gu. Self-play fine-tuning converts weak language models to strong language models. In International Conference on Machine Learning (ICML), 2024.

Chrisantha Fernando, Dylan Banarse, Henryk Michalewski, Simon Osindero, and Tim Rocktäschel. Promptbreeder: Self-referential self-improvement via prompt evolution. arXiv preprint arXiv:2309.16797, 2023a.

Chrisantha Fernando, Dylan Banarse, Henryk Michalewski, Simon Osindero, and Tim Rocktäschel. Promptbreeder: project page. https://github.com/google-deepmind/promptbreeder, 2023b.

Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. arXiv preprint arXiv:1503.02531, 2015.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive nlp tasks. In Advances in Neural Information Processing Systems (NeurIPS), 2020.

Xian Li, Ping Yu, Chunting Zhou, Timo Schick, Omer Levy, Luke Zettlemoyer, Jason Weston, and Mike Lewis. Self-alignment with instruction backtranslation. In International Conference on Learning Representations (ICLR), 2024.

Aman Madaan et al. Self-refine: Iterative refinement with self-feedback. In Advances in Neural Information Processing Systems (NeurIPS), 2023.

Nils Reimers and Iryna Gurevych. Sentence-bert: Sentence embeddings using siamese bert-networks. In Empirical Methods in Natural Language Processing (EMNLP), 2019.

Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Roberta Raileanu, Maria Lomeli, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. Toolformer: Language models can teach themselves to use tools. In Advances in Neural Information Processing Systems (NeurIPS), 2024.

Noah Shinn, Federico Cassano, Edward Berman, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. In Advances in Neural Information Processing Systems (NeurIPS), 2023.

Theodore R. Sumers, Shunyu Yao, Karthik Narasimhan, and Thomas L. Griffiths. Cognitive architectures for language agents. Transactions on Machine Learning Research (TMLR), 2024.

Rohan Taori, Ishaan Gulrajani, Tianyi Zhang, Yann Dubois, Xuechen Li, Carlos Guestrin, Percy Liang, and Tatsunori B. Hashimoto. Stanford alpaca: An instruction-following llama model. https://github.com/tatsu-lab/stanford_alpaca, 2023.

Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan, and Anima Anandkumar. Voyager: An open-ended embodied agent with large language models. arXiv preprint arXiv:2305.16291, 2023a.

Peiyi Wang, Lei Li, Liang Chen, Zefan Cai, Dawei Zhu, Binghuai Lin, Yunbo Cao, Qi Liu, Tianyu Liu, and Zhifang Sui. Position bias and fairness in large language model-as-a-judge. arXiv preprint arXiv:2403.xxxxx, 2024.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc V. Le, Ed H. Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In International Conference on Learning Representations (ICLR), 2023b.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H. Chi, Quoc V. Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In Advances in Neural Information Processing Systems (NeurIPS), 2022.

Can Xu, Qingfeng Sun, Kai Zheng, Xiubo Geng, Pu Zhao, Jiazhan Feng, Chongyang Tao, and Daxin Jiang. Wizardlm: Empowering large language models to follow complex instructions. International Conference on Learning Representations (ICLR), 2024.

Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Thomas L. Griffiths, Yuan Cao, and Karthik Narasimhan. Tree of thoughts: Deliberate problem solving with large language models. In Advances in Neural Information Processing Systems (NeurIPS), 2023a.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. React: Synergizing reasoning and acting in language models. In International Conference on Learning Representations (ICLR), 2023b.

Weizhe Yuan, Richard Yuanzhe Pang, Kyunghyun Cho, Sainbayar Sukhbaatar, Jing Xu, and Jason Weston. Self-rewarding language models. arXiv preprint arXiv:2401.10020, 2024.

Eric Zelikman, Yuhuai Wu, Jesse Mu, and Noah D. Goodman. Star: Bootstrapping reasoning with reasoning. In Advances in Neural Information Processing Systems (NeurIPS), 2022.

Eric Zelikman, Georges Harik, Yijia Shao, Varuna Jayasiri, Nick Haber, and Noah D. Goodman. Quiet-star: Language models can teach themselves to think before speaking. arXiv preprint arXiv:2403.09629, 2024.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P Xing, Hao Zhang, Joseph E Gonzalez, and Ion Stoica. Judging llm-as-a-judge with mt-bench and chatbot arena. In Advances in Neural Information Processing Systems (NeurIPS), 2023.

Pei Zhou, Jay Pujara, Xiang Ren, Xinyun Chen, Heng-Tze Cheng, Quoc V. Le, Ed H. Chi, Denny Zhou, Swaroop Mishra, and Huaixiu Steven Zheng. Self-discover: Large language models self-compose reasoning structures. In Advances in Neural Information Processing Systems (NeurIPS), 2024.

Xizhou Zhu et al. Ghost in the minecraft: Generally capable agents for open-world environments via large language models with text-based knowledge and memory. arXiv preprint arXiv:2305.17144, 2023.

A Wisdom record schema and example

Every wisdom is a JSON record:

```
{
  "id": "W077",
  "aphorism": "没有调查，就没有发言权",
  "source": "毛泽东《反对本本主义》",
  "signal": "多因混杂、观察噪声大、没定位清楚就已经要决策时",
  "unpacked_for_llm": "...",
  "cross_domain_examples": [
    {"domain": "software_engineering",
```

```
    "scenario": "多方反馈互相打架，先建复现与归因链路，再决定修固件还是改流程"},
   {"domain": "science",
    "scenario": "亮度异常成因多解且观测噪声大，先设计鉴别观测而非仓促定性"}
  ],
  "status": "active",
  "source": "success_distilled",
  "created_at": "2026-04-23T15:54:56Z",
  "keep_reason": "parallel validation wr_ext=0.60 on 50 holdout"
}
```


B Full prompt templates

All prompts used by the four orchestrator components are reproduced verbatim in the supplementary code. We list the section headers here for navigation:

- `success_distiller.py`: `DISTILL_PROMPT`
- `failure_generator.py`: `PROPOSE_PROMPT`
- `cross_llm_distiller.py`: `SOLVE_PROMPT`, `DISTILL_PROMPT`
- `validate_parallel.py`: `EXEMPLAR_PROMPT`
- `phase2_v20_framework.py`: `FRAME_REWRITE_PROMPT`, `EXECUTE_V20`, `REFLECT_PROMPT_V20`

C Additional evolution statistics

Table 2 expands on the three KEEP events. Table 3 lists the deprecated IDs.

ID	Aphorism	$\mathrm{wr}_{\mathrm{ext}}$	Source of signal	Provenance
W076	凡益之道，与时偕行	0.64	Turn-0 cluster $n = 9$	success_distilled
W077	没有调查，就没有发言权	0.60	Turn-0 cluster $n = 6$	success_distilled
W078	是骡子是马，拉出来遛遛	0.60	2 GPT-5.4-win math losses	cross_llm

Table 2: Three KEEP events this cycle.

Deprecated in v20.3 scan (0 activations across 5 selection files, 1000 top-2 slots): W005, W008, W014, W017, W018, W021, W027, W038, W053, W054, W055, W056, W061, W069, W071, W072, W073, W074, W075.

Table 3: The pruner's first-scan output.

D Compute and cost

Each v20 subprocess on 50 problems: ∼20 minutes wall time, ∼ input 180k tokens, output 40k tokens. Twelve parallel v20 runs in the validation gate: ∼45 minutes wall with 6 concurrent workers. Eleven parallel A/B judges (50 pairs each): ∼12 minutes with 8 concurrent judge threads. Total for one full success-distill → A/B cycle including exemplar mining: ∼1.5 hours wall time, within the budget of a single researcher's afternoon.